

DS-GA 1018: Homework 3

Due Monday October 28th at 5:00 pm

Problem 1 (19 points): Consider the latent space model we presented in class defined by:

$$\mathbf{z}_t = \mathbf{A}\mathbf{z}_{t-1} + \mathbf{w}_t \quad (1)$$

$$\mathbf{x}_t = \mathbf{C}\mathbf{z}_t + \mathbf{v}_t \quad (2)$$

where the latent space $\mathbf{z}$ has dimension d and the data $\mathbf{x}$ has dimension n . Our noise is being drawn from $\mathbf{w}_t \sim \mathcal{N}(0, \mathbf{Q})$ and $\mathbf{v}_t \sim \mathcal{N}(0, \mathbf{R})$. Assume that $n = d$ and that we have $\mathbf{A} = \mathbb{I}$, $\mathbf{C} = \mathbb{I}$, and $\mathbf{Q} = 0 \cdot \mathbb{I}$. Finally, assume that $\Sigma_0 \rightarrow \infty$. We make no specific assumption for $\mathbf{R}$ or μ_0 . We want to show that the smoothed mean $\mu_{t|T}$ simplifies to the mean of the data $\frac{1}{T} \sum_{i=1}^T \mathbf{x}_i$. This problem will walk you through the steps:

i. (5 points): Show that $\mu_{1|1} = \mathbf{x}_1$ and $\Sigma_{1|1} = \mathbf{R}$. You will need to take the limit $\Sigma_0 \rightarrow \infty$. *Hint: it will be useful to know that $(\mathbb{I} + \mathbf{M})^{-1} = \sum_{i=0}^{\infty} (-1)^i \mathbf{M}^i$, where $\mathbf{M}$ is an arbitrary matrix.*

We will compute the following quantities for preparations: $\mu_{0|0} = \mu_0$, $\Sigma_{0|0} = \Sigma_0$, $\mu_{1|0} = A\mu_{0|0} = A\mu_0$, $\Sigma_{1|0} = A\Sigma_{0|0}A^\top + Q = \Sigma_0$, $K_1 = \Sigma_{1|0}C^\top(C\Sigma_{1|0}C^\top + R)^{-1} = \Sigma_0(\Sigma_0 + R)^{-1}$. Then when $\Sigma_0 \rightarrow \infty$ we could get:

$$\begin{aligned} \mu_{1|1} &= \mu_{1|0} + K_1(x_1 - \mu_{1|0}) \\ &= \mu_0 + \Sigma_0(\Sigma_0 + R)^{-1}(x_1 - \mu_0) \\ &= \mu_0 + \Sigma_0(I + \Sigma_0^{-1}R)^{-1}\Sigma_0^{-1}(x_1 - \mu_0) \\ &= \mu_0 + \Sigma_0 \sum_{i=0}^{\infty} (-1)^i (\Sigma_0^{-1}R)^i \Sigma_0^{-1}(x_1 - \mu_0) \\ &= \mu_0 + \Sigma_0 \left(I + \sum_{i=1}^{\infty} (-1)^i (\Sigma_0^{-1}R)^i \right) \Sigma_0^{-1}(x_1 - \mu_0) \\ &\rightarrow \mu_0 + x_1 - \mu_0 = x_1 \end{aligned}$$

and

$$\begin{aligned}\Sigma_{1|1} &= \Sigma_{1|0} - K_1 C \Sigma_{1|0} \\ &= \Sigma_0 - \Sigma_0 (\Sigma_0 + R)^{-1} \Sigma_0 \\ &= \Sigma_0 - \Sigma_0 \left(\sum_{i=0}^{\infty} (-1)^i (\Sigma_0^{-1} R)^i \right) \Sigma_0^{-1} \Sigma_0 \\ &= \Sigma_0 - \Sigma_0 \left(I - \Sigma_0^{-1} R + \sum_{i=2}^{\infty} (-1)^i (\Sigma_0^{-1} R)^i \right) \\ &\rightarrow \Sigma_0 - \Sigma_0 + \Sigma_0 \Sigma_0^{-1} R + 0 \\ &\rightarrow R\end{aligned}$$

ii. (2 points): Explain the intuition behind the result for $\mu_{1|1}$ and $\Sigma_{1|1}$.

The intuition would be that since we have set the prior covariance to be $\Sigma_0 \rightarrow \infty$, we have little confidence in our hidden states and the next step prediction would solely depend on the data point. In other words, the step one covariance would totally depend on the covariance of the data point Σ_0 and the prediction would be exactly the data point x_1 . The fact that $K_1 \rightarrow I$ prove this intuition.

iii. (4 points): Assume that $\mu_{t-1|t-1} = \frac{1}{t-1} \sum_{i=1}^{t-1} \mathbf{x}_i$ and that $\Sigma_{t-1|t-1} = \frac{1}{t-1} \mathbf{R}$ (this is consistent with your previous result). Show that $\mu_{t|t} = \frac{1}{t} \sum_{i=1}^t \mathbf{x}_i$ and that $\Sigma_{t|t} = \frac{1}{t} \mathbf{R}$.

$$\text{We have } \mu_{t-1|t-1} = \frac{1}{t-1} \sum_{i=1}^{t-1} x_i \text{ and } \sum_{t-1|t-1} = \frac{1}{t-1} R$$

$$\therefore \mu_{t|t-1} = A\mu_{t-1|t-1} = \frac{1}{t-1} \sum_{i=1}^t x_i$$

$$\Sigma_{t|t-1} = A\Sigma_{t|t-1}A^\top + Q = \frac{1}{t-1} R$$

$$\begin{aligned} K_t &= \Sigma_{t|t-1}C^\top \left(C\Sigma_{t|t-1}C^\top + R \right) \\ &= \frac{1}{t-1} \cdot R \left(\frac{1}{t-1} R + R \right)^{-1} \\ &= \frac{1}{t-1} \cdot R \frac{t-1}{t} R^{-1} \\ &= \frac{1}{t} \end{aligned}$$

$$\begin{aligned} \mu_{t|t} &= \mu_{t|t-1} + K_t (X_t - \mu_{t|t-1}) \\ &= \frac{1}{t-1} \sum_{i=1}^{t-1} x_i - \frac{1}{t(t-1)} \sum_{i=1}^{t-1} x_i + \frac{1}{t} x_t \\ &= \frac{1}{t} \sum_{i=1}^{t-1} x_i + \frac{1}{t} x_t \\ &= \frac{1}{t} \sum_{i=1}^t x_i \end{aligned}$$

$$\begin{aligned} \Sigma_{t|t} &= \Sigma_{t|t-1} - K_t C \Sigma_{t|t-1} \\ &= \frac{1}{t-1} R - \frac{1}{t} \cdot \frac{1}{t-1} R \\ &= \left(\frac{1}{t-1} - \frac{1}{t-1} + \frac{1}{t} \right) R \\ &= \frac{1}{t} \cdot R \end{aligned}$$

iv. (6 points): Finally, show that $\boldsymbol{\mu}_{t|T} = \frac{1}{T} \sum_{i=1}^T \mathbf{x}_i$ and $\boldsymbol{\Sigma}_{t|T} = \frac{1}{T} \mathbf{R}$ for all t .

We have the following:

$$\begin{aligned}\mu_{t|t} &= \frac{1}{t} \sum_{i=1}^t x_i \\ \Sigma_{t|t} &= \frac{1}{t} R \\ \mu_{t+1|t} &= A \cdot \frac{1}{t} \sum_{i=1}^t x_i = \frac{1}{t} \sum_{i=1}^t x_i \\ \Sigma_{t+1|t} &= \frac{1}{t} R + Q = \frac{1}{t} R \\ F_t &= \Sigma_{t|t} A^\top \Sigma_{t+1|t}^{-1} \\ &= \frac{1}{t} R \cdot \left(\frac{1}{t} R \right)^{-1} \\ &= I\end{aligned}$$

So we know:

$$\begin{aligned}\text{Assume } \mu_{t+1|t} &= \frac{1}{T} \sum_{i=0}^T x_i, \Sigma_{t+1|T} = \frac{1}{T} R \\ \mu_{t|T} &= \mu_{t|t} + F_t (\mu_{t+1|T} - \mu_{t+1|t}) \\ &= \frac{1}{t} \sum_{i=1}^t x_i + \frac{1}{T} \sum_{i=0}^T x_i - \frac{1}{t} \sum_{i=1}^t x_i \\ &= \frac{1}{T} \sum_{i=0}^T x_i \\ \Sigma_{t|T} &= F_t (\Sigma_{t+1|T} - \Sigma_{t+1|t}) F_t^\top + \Sigma_{t|t} \\ &= \left(\frac{1}{T} R - \frac{1}{t} R \right) + \frac{1}{t} R \\ &= \frac{1}{T} R\end{aligned}$$

By induction, the statement is proved.

v. (2 points): Explain the intuition behind the final result.

1. **Mean** $\mu_{t|T}$: In a system with no process noise and independent observations, the best estimate of the state at time t is the average of all the observations. Each observation contributes equally to the estimate, which is why the estimate is simply the arithmetic mean of the data points x_i .
2. **Covariance** $\Sigma_{t|T}$: As we incorporate more observations into the estimate, the uncertainty in the estimate decreases. Each observation has noise covariance R , and when we average T independent noisy measurements, the resulting uncertainty is reduced by a factor of T . Thus, the covariance of the estimate decreases as T increases, becoming $\frac{1}{T}R$.

Problem 2 (20 points): In lecture we discussed the importance of the initial parameter values for the results of the expectation-maximization algorithm. In this problem we will quantitatively demonstrate that importance. Consider the EM algorithm for an LDS system like the one in class. Imagine that we have an initial guess for our parameters:

$$\boldsymbol{\theta}_0 = \{\boldsymbol{\mu}_{0,0}, \boldsymbol{\Sigma}_{0,0}, \mathbf{A}_0, \mathbf{Q}_0, \mathbf{C}_0, \mathbf{R}_0\}. \quad (3)$$

We have also set our distribution $q_1(\mathbf{Z}) = p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta}_0)$. In other words, q_1 is the distributions set by the Kalman smoothing means and covariances (the standard choice).

i. (3 points): Consider the case where $\mathbf{A}_0 = 0 \cdot \mathbb{I}$ and $\mathbf{C}_0 = 0 \cdot \mathbb{I}$. what will the value of $\boldsymbol{\mu}_{0,1}$ be in terms of the data in $\mathbf{X}$ and the parameters in $\boldsymbol{\theta}_0$? For all the subproblems, don't forget to consider how our choice of $\boldsymbol{\theta}_0$ affects terms like $\boldsymbol{\mu}_{t|t-1}$.

We know that if $A_0 = 0$ and C_0 , then the filtering step will be:

1. $\mu_{t|t-1} = A_0 \mu_{t-1|t-1} = 0$
2. $\Sigma_{t|t-1} = A_0 \Sigma_{t-1|t-1} A_0^\top + Q_0 = Q_0$
3. $K_t = \Sigma_{t|t-1} C_0^\top (C_0 \Sigma_{t|t-1} C_0^\top + R)^{-1} = 0$
4. By induction, we know $\mu_{t|t} = 0, \forall t = 0, 2, \dots, T$ and $\Sigma_{t|t} = Q_0, \forall t = 0, 1, \dots, T$

and the smoothing step would be:

1. $F_t = \Sigma_{t|t} A_0^\top \Sigma_{t+1|t}^{-1} = 0, \forall t = 0, 1, \dots, T$
2. $\mu_{t|T} = \mu_{t|t} + 0 \cdot (\mu_{t+1|T} - \mu_{t+1|t}) = 0, \forall t = 0, 1, \dots, T$
3. $\Sigma_{t|T} = F_t (\Sigma_{t+1|T} - \Sigma_{t+1|t}) F_t^\top + \Sigma_{t|t} = Q_0, \forall t = 0, 1, \dots, T$

thus we have

1. $\mu_{0,1} = \mu_{0|T} = 0$

ii. (3 points): Still considering the case where $\mathbf{A}_0 = 0 \cdot \mathbb{I}$ and $\mathbf{C}_0 = 0 \cdot \mathbb{I}$, what will the value of $\boldsymbol{\Sigma}_{0,1}$ be in terms of the data in $\mathbf{X}$ and the parameters in $\boldsymbol{\theta}_0$?

By the derivations in (i) we have $\Sigma_{0,1} = \mathbb{E}[z_0 z_0^T] - \boldsymbol{\mu}_{0,n} \boldsymbol{\mu}_{0,n}^T = \Sigma_{0|T} = Q_0$.

iii. (4 points): Still considering the case where $\mathbf{A}_0 = 0 \cdot \mathbb{I}$ and $\mathbf{C}_0 = 0 \cdot \mathbb{I}$, what will the value of $\mathbf{A}_1$ be in terms of the data in $\mathbf{X}$ and the parameters in $\boldsymbol{\theta}_0$?

$\mathbb{E}_{q_n} [\mathbf{z}_t \mid \mathbf{x}_{1:T}] = \boldsymbol{\mu}_{t|T}$

We know $\mathbb{E}_{q_n} [\mathbf{z}_t \mathbf{z}_t^T \mid \mathbf{x}_{1:T}] = \boldsymbol{\Sigma}_{t|T} + \boldsymbol{\mu}_{t|T} \boldsymbol{\mu}_{t|T}^T$ so combined

$\mathbb{E}_{q_n} [\mathbf{z}_{t+1} \mathbf{z}_t^T \mid \mathbf{x}_{1:T}] = \mathbf{F}_t \boldsymbol{\Sigma}_{t+1|T} + \boldsymbol{\mu}_{t+1|T} \boldsymbol{\mu}_{t|T}^T$.

with (i) we will have:

1. $\mathbb{E}[\mathbf{z}_t] = \boldsymbol{\mu}_{0|T} = 0, \forall t = 0, 1, \dots, T$
2. $\mathbb{E}[\mathbf{z}_t \mathbf{z}_{t-1}^T] = \mathbf{F}_{t-1} \boldsymbol{\Sigma}_{t|T} + \boldsymbol{\mu}_{t|T} \boldsymbol{\mu}_{t-1|T}^T = 0, \forall t = 1, 2, \dots, T$
3. $\mathbb{E}[\mathbf{z}_t \mathbf{z}_t^T] = \boldsymbol{\Sigma}_{t|T} + \boldsymbol{\mu}_{t|T} \boldsymbol{\mu}_{t|T}^T = \mathbf{Q}_0 + \boldsymbol{\mu}_{0|T} \boldsymbol{\mu}_{0|T}^T = \mathbf{Q}_0, \forall t = 0, 1, \dots, T$

Thus we know $\mathbf{A}_1 = \left(\sum_{t=1}^T \mathbb{E} [\mathbf{z}_t \mathbf{z}_{t-1}^T] \right) \left(\sum_{t=1}^T \mathbb{E} [\mathbf{z}_{t-1} \mathbf{z}_{t-1}^T] \right)^{-1} = 0 \cdot \left(\sum_{t=1}^T \mathbb{E} [\mathbf{z}_{t-1} \mathbf{z}_{t-1}^T] \right)^{-1} = 0 \cdot \mathbb{I}$

iv. (4 points): Still considering the case where $\mathbf{A}_0 = 0 \cdot \mathbb{I}$ and $\mathbf{C}_0 = 0 \cdot \mathbb{I}$, what will the value of $\mathbf{Q}_1$ be in terms of the data in $\mathbf{X}$ and the parameters in $\boldsymbol{\theta}_0$?

We know that $\mathbf{Q}_1 = \frac{1}{T} \left(\sum_{t=1}^T \mathbb{E} [\mathbf{z}_t \mathbf{z}_t^T] - \mathbf{A}_n \mathbb{E} [\mathbf{z}_{t-1} \mathbf{z}_t^T] - \mathbb{E} [\mathbf{z}_t \mathbf{z}_{t-1}^T] \mathbf{A}_n^T + \mathbf{A}_n \mathbb{E} [\mathbf{z}_{t-1} \mathbf{z}_{t-1}^T] \mathbf{A}_n^T \right)$

Based on the derivations in (iii) we have $\mathbf{Q}_1 = \frac{1}{T} (\sum_{i=1}^T \mathbf{Q}_0 - 0 - 0 + 0) = \mathbf{Q}_0$

v. (4 points): Still considering the case where $\mathbf{A}_0 = 0 \cdot \mathbb{I}$ and $\mathbf{C}_0 = 0 \cdot \mathbb{I}$, what will the value of $\mathbf{C}_1$ be in terms of the data in $\mathbf{X}$ and the parameters in $\boldsymbol{\theta}_0$?

We know from previous derivations that $\mathbf{C}_1 = \left(\sum_{t=1}^T \mathbf{x}_t \mathbb{E} [\mathbf{z}_t^T] \right) \left(\sum_{t=1}^T \mathbb{E} [\mathbf{z}_t \mathbf{z}_t^T] \right)^{-1} = (\sum_{t=1}^T \mathbf{x}_t \cdot 0) (\mathbf{Q}_0)^{-1} = 0 \cdot \mathbb{I}$.

vi. (2 points): Given these results, what is the solution to $\mu_{0,n}$, $\Sigma_{0,n}$, A_n , C_n , and Q_n ?

We know from that if $A_n = 0 \cdot \mathbb{I}$, $C_n = 0 \cdot \mathbb{I}$, $Q_n = Q_0$, $C_n = 0 \cdot \mathbb{I}$, then from the previous derivations we know $\mu_{0,n+1} = \mu_{0|T} = 0$, $\Sigma_{0,n+1} = \Sigma_{0|T} = Q_n$, $A_{n+1} = 0 \cdot \mathbb{I}$, $C_{n+1} = 0 \cdot \mathbb{I}$, and $Q_{n+1} = Q_n = Q_0$. So by induction, we have

1. $\mu_{0,n} = \mu_{0|T} = 0$
2. $\Sigma_{0,n} = \Sigma_{0|T} = Q_0$
3. $A_n = 0 \cdot \mathbb{I}$
4. $Q_n = Q_0$

for all $n = 0, 1, \dots$